Abolfazl Ziaeemehr

Research engineer, scientific software and machine learning

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Profile

I build fast, differentiable software for simulating dynamical systems and fitting them to data. At Aix-Marseille University I led the development of VBI, a toolkit for simulation-based inference on whole-brain models, and in 2026 I wrote and released JaxCont and lyapax (numerical analysis in JAX) and LitForge (a local-first literature app). I care about tested code, documentation and releases that other people can install.

Experience

Oct 2026–now **Research engineer (Ingénieur de recherche)**, *Université de Picardie Jules Verne*, Amiens, France

Oct 2021–Sep 2026 **Postdoctoral researcher**, *Institut de Neurosciences des Systèmes, Inserm UMR 1106, Aix-Marseille University*, Marseille, France

With Viktor Jirsa and Meysam Hashemi.

- Lead developer of VBI: C++, CuPy and Numba simulators, feature extraction and PyTorch neural density estimators for inferring brain-model parameters from EEG, MEG, SEEG and fMRI.
- Applied simulation-based inference in studies of epilepsy, multiple sclerosis, Parkinson's disease, Alzheimer's disease and schizophrenia.
- Contributor to the *sbi* library; taught high-performance Python to the group.

Mar–Sep 2021 **Senior scientific developer**, *Panoptopia*

Python packages for costing and risk management.

Sep 2020–Mar 2021 **Researcher**, *Institute for Research in Fundamental Sciences (IPM)*, Tehran

Spiking and rate models of the basal ganglia in Parkinson's disease.

Apr 2019–Feb 2020 **Research assistant**, *IPM*, Tehran

Chimera states and population models on the human connectome.

Apr–Sep 2018 **Visiting researcher**, *University of Granada, Computational Physics Group*, Spain

Phase transitions and noise-induced volatility in networks of integrate-and-fire neurons.

Software

VBI Virtual Brain Inference, lead developer. Simulation-based inference for whole-brain models; published in eLife (2025) and JOSS (2026). PyPI, EBRAINS.

JaxCont Differentiable continuation and bifurcation analysis in JAX: equilibrium and periodic-orbit branches, fold, Hopf, branch-point and period-doubling detection, two-parameter curves; compiles with `jit`, batches with `vmap`, differentiates with `grad`. Validated against MatCont 7.6. PyPI, Read the Docs, Zenodo.

lyapax Lyapunov exponent spectra of ODEs, maps and delay equations in JAX via forward-mode tangent propagation; runs on GPU; benchmarked against ChaosTools.jl. PyPI, Read the Docs.

LitForge Local-first desktop app for literature discovery and screening, question answering with verified quotes, and LaTeX writing with local or cloud language models.

MD Viewer — Bidirectional Persian and English Markdown viewer with KaTeX, released for web, desktop (Tauri) and Android.

Other — Contributions to sbi, NESTML and Brian2; JITCSIM, vbjax, mndynamics; reproducible code for each of my first-author papers. Full list at ziaeemehr.github.io.

Skills

Languages — Python, C++, JavaScript, Julia, Bash

Numerics — JAX, NumPy, SciPy, Numba, CuPy and CUDA; ODE, SDE and DDE solvers, continuation and bifurcation methods, Lyapunov analysis

Machine learning — PyTorch, simulation-based inference (sbi), normalising flows and neural density estimators, CNNs for EEG

Data — EEG, MEG, SEEG and fMRI; MNE-Python, The Virtual Brain, NEST, Brian2

Engineering — Git, GitHub Actions, Linux, PyPI packaging and releases, pytest, Sphinx and Read the Docs, SWIG bindings, Tauri, HPC clusters

Education

2014–2020 — **PhD, computational neuroscience**, *Institute for Advanced Studies in Basic Sciences (IASBS)*, Zanjan, Iran
Thesis: *Synchronization dynamics on undirected and directed hierarchical networks*. Supervisors: Mina Zarei and Alireza Valizadeh.

2008–2011 — **MSc, solid-state physics**, *Imam Khomeini International University*, Qazvin, Iran
Thesis: *Calculation of optical properties of metallic nanoparticles*.

2004–2008 — **BSc, physics**, *University of Qom*, Iran

Publications

2026 — G. Preti, H. E. Wang, **A. Ziaeemehr**, et al. **Virtual brain twins guide personalized treatment decision in schizophrenia**. *medRxiv*, 2026.

2026 — A. Esmaeili, M. Woodman, N. Baldy, **A. Ziaeemehr**, et al. **Cohort-amortized personalization: navigating the privacy-utility frontier for virtual brain twins**. *arXiv*, 2026.

2026 — **A. Ziaeemehr**, L. Domide, M. Woodman, S. Petkoski, V. Jirsa, M. Hashemi. **Virtual Brain Inference (VBI) toolkit on EBRAINS**. *Journal of Open Source Software* 11(124), 10612, 2026.

2026 — M. Hashemi, D. Depannemaecker, M. Saggio, ..., **A. Ziaeemehr**, et al. **Principles and operation of virtual brain twins**. *IEEE Reviews in Biomedical Engineering* 19, 111–139, 2026.

2025 — **A. Ziaeemehr**, M. Woodman, L. Domide, S. Petkoski, V. Jirsa, M. Hashemi. **Virtual Brain Inference (VBI), a flexible and integrative toolkit for efficient probabilistic inference on whole-brain models**. *eLife* 14, RP106194, 2025.

2025 — A. Esmaeili, M. Demolliens, M. Viersen, **A. Ziaeemehr**, et al. **Probabilistic inference of social presence across brain scales reveals enhanced synaptic efficacy**. *Communications Biology* 8, 1608, 2025.

2025 — J. Boelts, M. Deistler, M. Gloeckler, ..., **A. Ziaeemehr**, ..., J. H. Macke. **sbi reloaded: a toolkit for simulation-based inference workflows**. *Journal of Open Source Software* 10(108), 7754, 2025.

2025 — G. Rabuffo, H. Lokossou, Z. Li, **A. Ziaeemehr**, et al. **Mapping global brain reconfigurations following local targeted manipulations**. *PNAS* 122(16), e2405706122, 2025.

2025 C. Mazzara, **A. Ziaeemehr**, E. Troisi Lopez, et al. **Mapping brain lesions to conduction delays: the next step for personalized brain models in multiple sclerosis**. *Human Brain Mapping* 46(7), e70219, 2025.

2025 M. Angiolelli, D. Depannemaecker, H. Agouram, ..., **A. Ziaeemehr**, et al. **The virtual parkinsonian patient**. *npj Systems Biology and Applications* 11, 40, 2025.

2024 M. Hashemi, **A. Ziaeemehr**, M. Woodman, J. Fousek, S. Petkoski, V. Jirsa. **Simulation-based inference on virtual brain models of disorders**. *Machine Learning: Science and Technology* 5(3), 035019, 2024.

2024 P. Sorrentino, A. Pathak, **A. Ziaeemehr**, et al. **The virtual multiple sclerosis patient**. *iScience* 27(7), 110101, 2024.

2023 B. H. Yalçınkaya, **A. Ziaeemehr**, J. Fousek, et al. **Personalized virtual brains of Alzheimer's disease link dynamical biomarkers of fMRI with increased local excitability**. *medRxiv*, 2023.

2021 **A. Ziaeemehr**, A. Valizadeh. **Frequency-resolved functional connectivity: role of delay and the strength of connections**. *Frontiers in Neural Circuits* 15, 608655, 2021.

2020 **A. Ziaeemehr**, M. Zarei, A. Valizadeh, C. R. Mirasso. **Frequency-dependent organization of the brain's functional network through delayed-interactions**. *Neural Networks* 132, 155–165, 2020.

2020 **A. Ziaeemehr**, M. Zarei, A. Sheshbolouki. **Emergence of global synchronization in directed excitatory networks of type I neurons**. *Scientific Reports* 10, 3306, 2020.

Teaching

2026 Deep learning by doing: self-study course from tensors to CNNs for EEG classification.

2024 High-performance computing with Python, workshop for the Theoretical Neuroscience Group, INS.

Jul 2020 Teaching assistant, Neuromatch Academy (three-week summer school).

2016–2017 Workshop lecturer at IASBS: Python scripting, Julia, C++, Brian2 and NEST.

2015–2016 Teaching assistant: classical electrodynamics I and II, computational physics.

Awards

2018 Ministry of Science of Iran scholarship for research at the University of Granada, Spain.

2014 Ranked 26th of about 5,000 candidates in the national PhD entrance exam.

Languages

Persian Native

English Fluent

French Intermediate

References

Viktor Jirsa, Director of Research, INS, Aix-Marseille University
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